SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning

Data Visualization Lab

Course Code: TE7663

Assignment Practical No. 7

Integrating R with Power BI: Time Series Analysis of Airline Passengers

A Blog-Style Report

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Introduction

Business Intelligence tools are excellent at summarizing data, but they become far more powerful when paired with a statistical programming language. In this practical, I combined Microsoft Power BI Desktop with R to build an interactive dashboard around the classic AirPassengers dataset, which records monthly airline passenger counts over many years. The advanced analytics technique chosen for this use case is time series analysis, covering trend estimation, seasonality exploration and forecasting.

The goal of this blog is to walk through every step I performed, from configuring R inside Power BI to creating KPI cards and R-based visuals, and finally to interpret what the data tells us from a business point of view.

Step 1: Configuring R with Power BI

The first task was to install R on the system and point Power BI Desktop to the R home directory through *File > Options and settings > Options > R scripting*. Once Power BI detected the R installation, R scripts could be executed both as data sources and as visuals. The ggplot2 package was installed in R so that it could be used for all the custom charts in the dashboard.

Step 2: Importing the Dataset

The AirPassengers data was loaded into Power BI as a table named **AirPassengers_df**. A second table named **Forecast** was also added to hold the projected passenger values generated by the time series model. Both tables are visible in the Data pane of the report, as shown below.

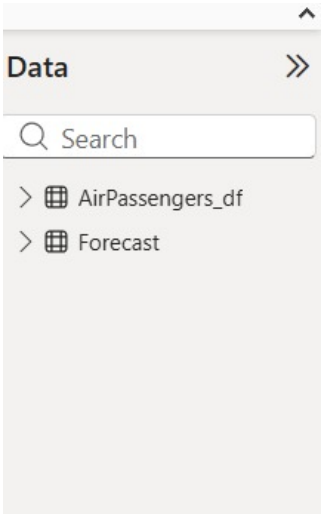


Figure 1: Data pane showing the AirPassengers_df and Forecast tables

Step 3: Data Cleaning and Preparation

Before analysis, the data was checked for missing values and duplicate rows. Power BI automatically removes duplicate rows when an R script is used (as the R script editor warns), and any missing values were handled in Power Query. The dataset was then enriched with additional columns such as Year, Quarter, Month, MonthNum, a Date field and a Time Hierarchy. The hierarchy of Year, Quarter, Month and Day is what enables drill-down in the visuals.

A set of DAX measures was also created in the model to power the KPI cards: **Total Passengers**, **Avg Monthly Passengers**, **Peak Month Passengers**, **Lowest Month Passengers** and **YoY Growth %**. These five measures satisfy the minimum requirement of five KPI cards on the dashboard.

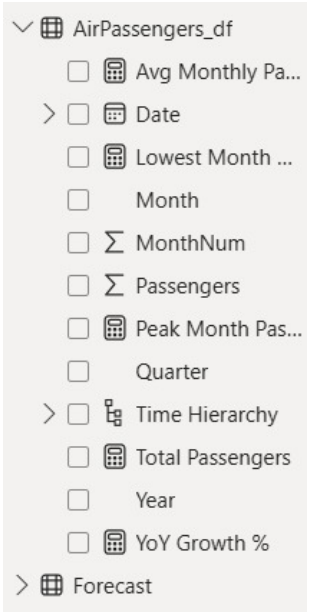


Figure 2: Fields of AirPassengers_df, including the calculated measures used as KPI cards

Step 4: Writing and Executing the R Script

With the data prepared, the next step was to add an R script visual to the report canvas. Power BI passes the selected fields to the script as a data frame called *dataset*, with duplicates already removed. The first R visual is a line chart of monthly passengers over time, with a LOESS smoothing curve overlaid as a dashed red line to expose the long-term trend. The data is first sorted by date so that the line is drawn in the correct order.

R script editor

⚠ Duplicate rows will be removed from the data.

```r
# The following code to create a dataframe and remove duplicated rows is always executed

# dataset <- data.frame(Year, Quarter, Month, Day, Passengers)
# dataset <- unique(dataset)

# Paste or type your script code here:
library(ggplot2)
df <- dataset[order(dataset$Date), ]
ggplot(df, aes(Date, Passengers)) +
  geom_line(color = "#1f77b4", linewidth = 1) +
  geom_smooth(method = "loess", se = FALSE, color = "red", linetype = "dashed") +
  labs(title = "Monthly Air Passengers with Trend", x = "Year", y = "Passengers") +
  theme_minimal(base_size = 12)
```

Figure 3: R script editor in Power BI with the ggplot2 code for the passenger trend line

This script ran successfully and produced the Monthly Air Passengers with Trend visual. The solid line shows the raw monthly values, while the dashed LOESS curve smooths out short-term noise and clearly reveals the steady upward direction of air travel demand.

Step 5: Summary Statistics

Descriptive statistics were generated to understand the spread of the data. The mean, median, minimum, maximum and quartiles of the Passengers column give a quick picture of the typical monthly demand and its variability. In the dashboard these figures are surfaced through five KPI cards: Total Passengers of 40.36K, Average Monthly Passengers of 280.3, Lowest Month Passengers of 104, Peak Month Passengers of 622, and Year-over-Year Growth of 11.17%.

Step 6: Seasonality Visualization in R

To study seasonality, a second R visual was built. The script converts the Year and MonthNum columns into a proper Date, extracts the abbreviated month name as an ordered factor, and then draws a boxplot of passengers for each month using ggplot2. Setting the month as an ordered factor ensures that the x-axis runs from January to December rather than alphabetically.

```r
library(ggplot2)
dataset$Date <- as.Date(paste(dataset$Year, dataset$MonthNum, 1, sep = "-"))
dataset$MonthName <- factor(format(dataset$Date, "%b"), levels = month.abb)
ggplot(dataset, aes(MonthName, Passengers, fill = MonthName)) +
  geom_boxplot(show.legend = FALSE) +
  labs(title = "Seasonality: Passengers by Month", x = "Month", y = "Passengers") +
  theme_minimal(base_size = 12)

```

Figure 4: R code used to build the seasonality boxplot

The resulting chart, titled *Seasonality: Passengers by Month*, uses a different fill colour for each month and the minimal theme for a clean, professional appearance.

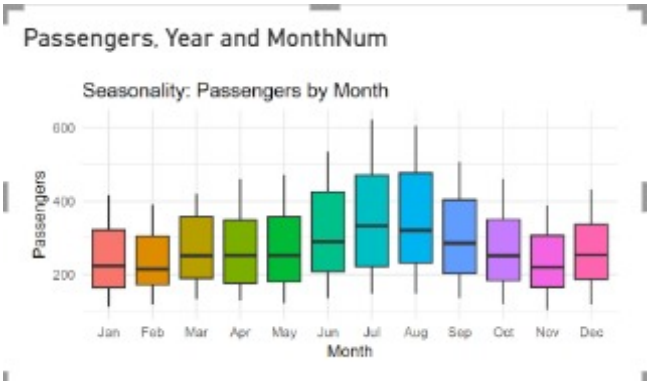


Figure 5: Seasonality boxplot of passengers by month

Step 7: Advanced Analytics

The statistical technique implemented in this practical is time series analysis, carried out in three parts. First, the trend was estimated with LOESS (locally estimated scatterplot smoothing), which fits a smooth curve through the series without assuming a rigid functional form. Second, a time series decomposition split the series into its observed, trend, seasonal and remainder components, making the repeating yearly pattern easy to see. Third, a Holt-Winters model was fitted to forecast the next 24 months. The widening gap between summer highs and winter lows as the trend rises indicates a multiplicative seasonal pattern, so the multiplicative Holt-Winters version was the best fit. The forecast values are stored in the Forecast table and drawn as a red line continuing from the historical data.

Step 8: Building the Interactive Dashboard

The R visuals were placed on the same page as native Power BI visuals so that the dashboard works as a single analytical view. The five KPI cards, each in a different colour, run across the top of the page. Beside them sits a Year slicer covering 1949 to 1960, and Quarter and Month slicers sit on the right-hand side. Below the cards, the page holds four R visuals: the monthly passenger trend with a LOESS line, the seasonality boxplot, the time series decomposition and the 24-month Holt-Winters forecast. It also holds four native Power BI visuals: a donut chart of passengers by quarter, a matrix of passengers by year and month, a line chart by year and a column chart by year. A Key Insights text box at the bottom right summarizes the main findings.

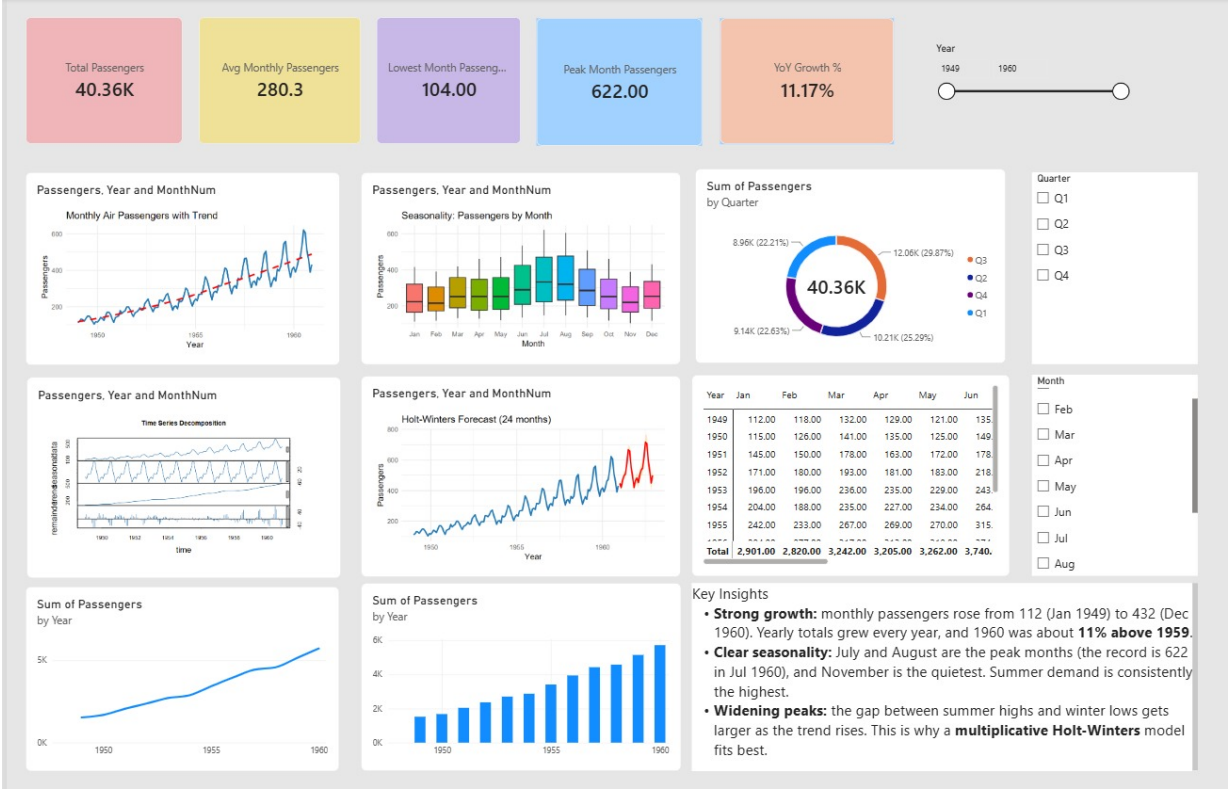


Figure 6: Final interactive Power BI dashboard with KPI cards, slicers, R visuals and Power BI visuals

The slicers and filters apply to the whole page, and the Year, Quarter and Month hierarchy allows drill-down in the Power BI visuals. Because R visuals respond to Power BI filters, changing a slicer redraws the ggplot2 charts instantly, which is what makes the dashboard truly interactive.

Step 9: Business Insights

Strong growth. Monthly passengers rose from 112 in January 1949 to 432 in December 1960. Yearly totals grew every year, and 1960 was about 11% above 1959, matching the 11.17% YoY Growth card. Total traffic across the period is 40.36K passengers.

Clear seasonality. July and August are the peak months, with the record of 622 passengers in July 1960, while November is the quietest. Summer demand is consistently the highest. The quarterly donut agrees, with Q3 contributing the largest share at 29.87% and Q1 the smallest at 22.21%.

Widening peaks. The gap between summer highs and winter lows gets larger as the trend rises. This is why the multiplicative Holt-Winters model fits best, and it also means the busy months are less predictable, which matters for capacity planning.

Forecast. The 24-month forecast continues both the upward trend and the summer peaks, suggesting that demand will keep growing.

Recommendations. Airlines can add capacity and staff before the summer peak, run promotional pricing in the low season to improve load factors, and use the forecast to plan fleet scheduling and budgets.

Step 10: Conclusion

This practical showed how R and Power BI complement each other. Power BI provided the interactive layer with KPI cards, slicers, filters and drill-down, while R supplied the statistical depth through ggplot2-based trend and seasonality visuals. The analysis of the AirPassengers dataset revealed a clear long-term growth trend, a strong summer seasonality and a forecast of continued growth, and these insights can directly support operational and commercial decisions. Overall, integrating R with Power BI turns a standard dashboard into a richer analytical tool.

Tools used: Microsoft Power BI Desktop, R, ggplot2, DAX